OPENSOURCESDRLAB DOCUMENTATION

# FMC-AD9371 Schematic

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| Product | FMC-AD9371 |
| Document | OSSDRLAB-FMC-AD9371-SCH |
| Version | 1.0 |
| Date | 2026-11-12 |

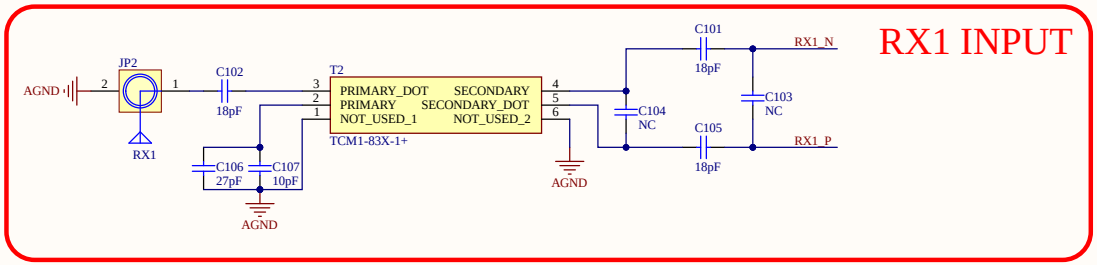
This document contains the circuit schematic of the OpenSourceSDRLab FMC-AD9371 board, exported from Altium Designer (A3 and A4 sheets). The seven schematic sheets follow this cover page.

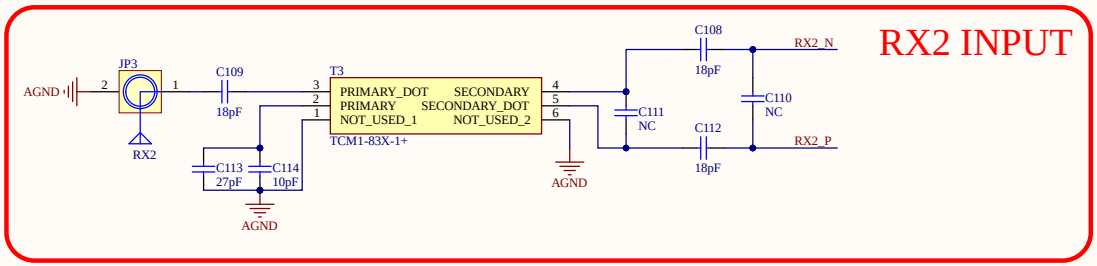
| Sheet | Content |
|---|---|
| 1 | RF inputs: RX1, RX2, ORX1, ORX2 baluns |
| 2 | AD9371 RF pins, RF outputs TX1 and TX2, RX and TX external LO input/output |
| 3 | Clocking: AD9528 clock generator, reference clock input, 122.88 MHz VCXO, status LEDs, SPI read-back buffer, level translators, clock supply filtering |
| 4 | AD9371 device clock, SPI, control and GPIO pins; JESD204B SERDES lanes, SYNC and SYSREF |
| 5 | AD9371 power pins and decoupling |
| 6 | Power supply: 12 V input, ADP5054 regulator, feedback, filtering, rail enables, interface and clock supplies |
| 7 | FMC HPC connector and ID EEPROM |

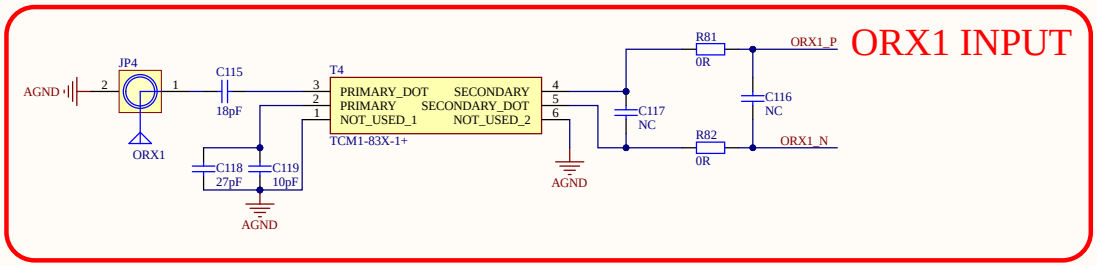
Product page: https://opensourcesdrlab.com/products/opensourcesdrlab-high-performance-ad9371-fmc-board-for-fpga-mezzanine-card-compatible-with-official-version

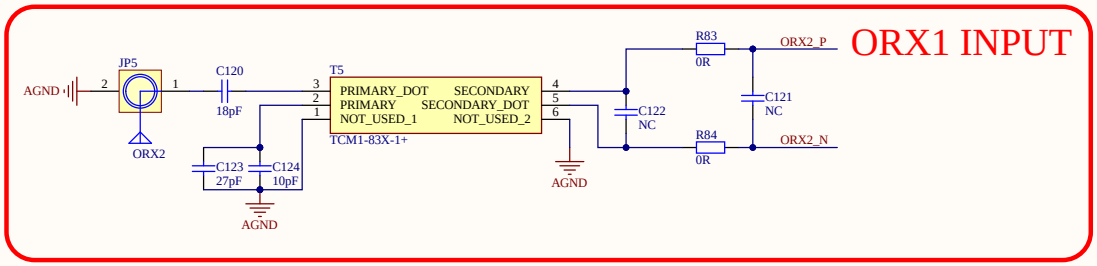
Questions about the hardware: support@opensourcesdrlab.com

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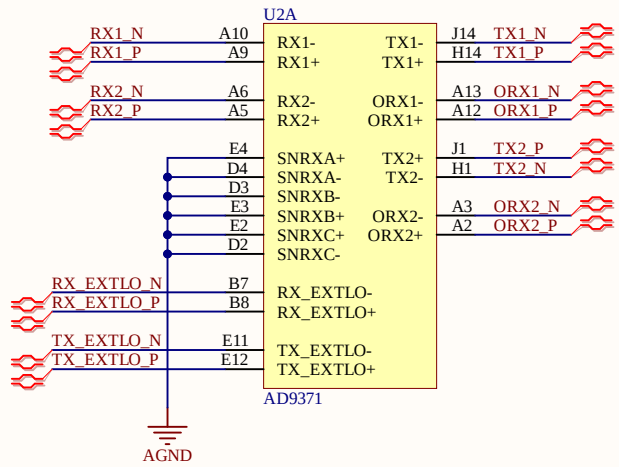


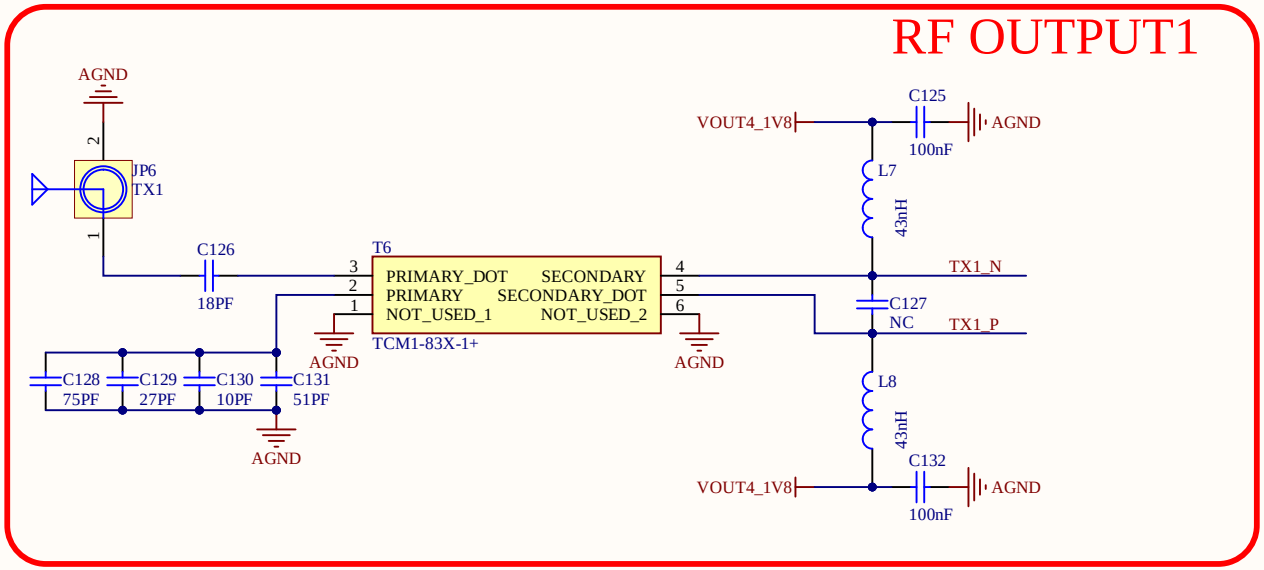



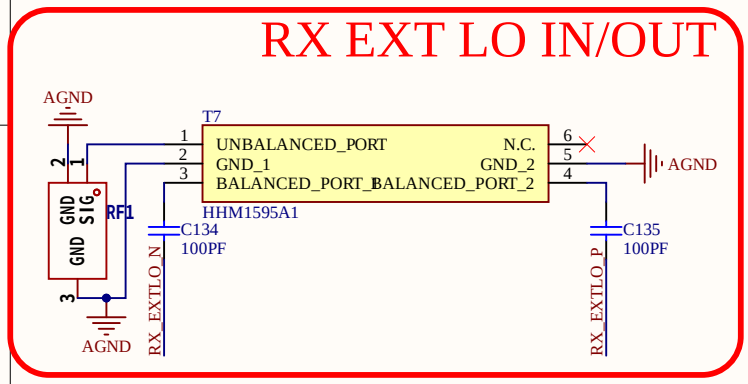



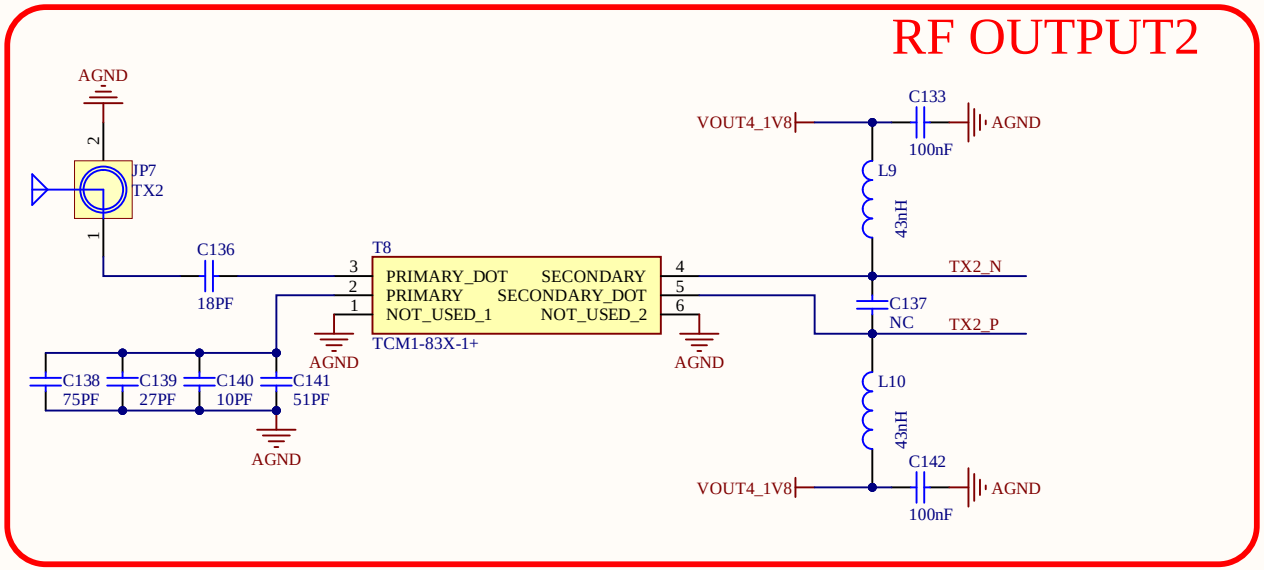



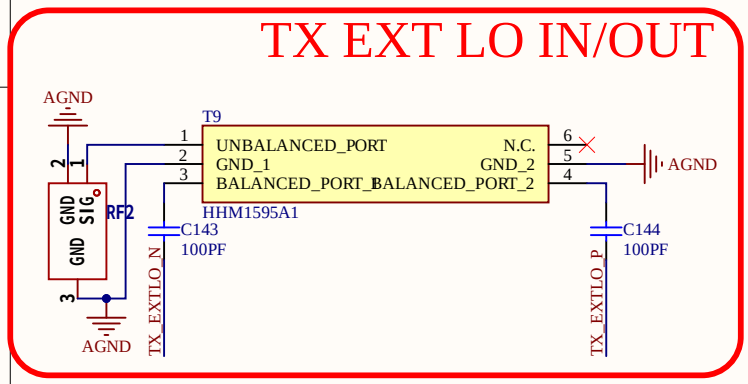

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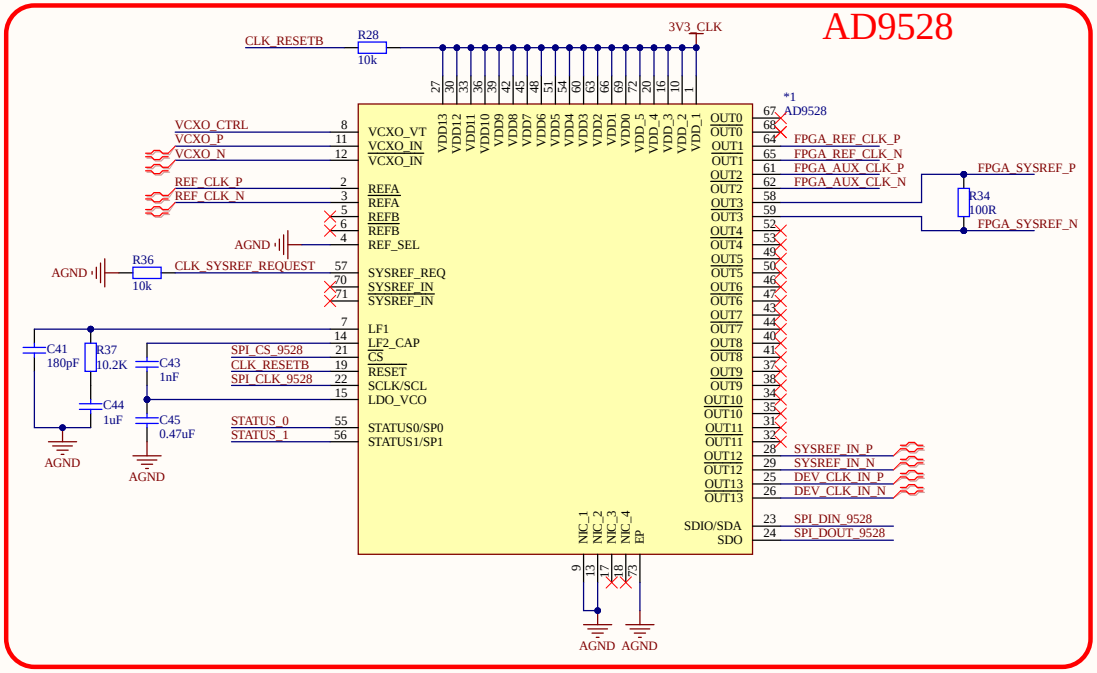



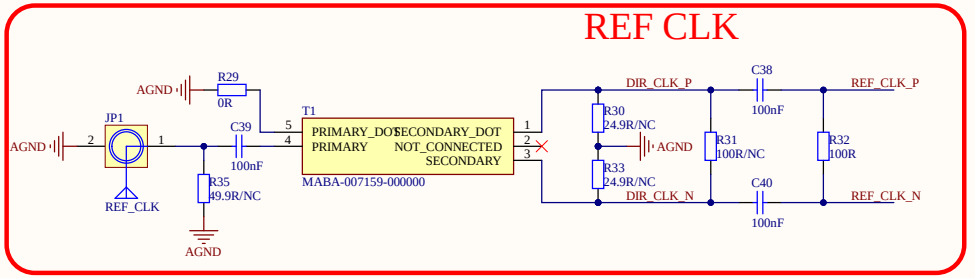



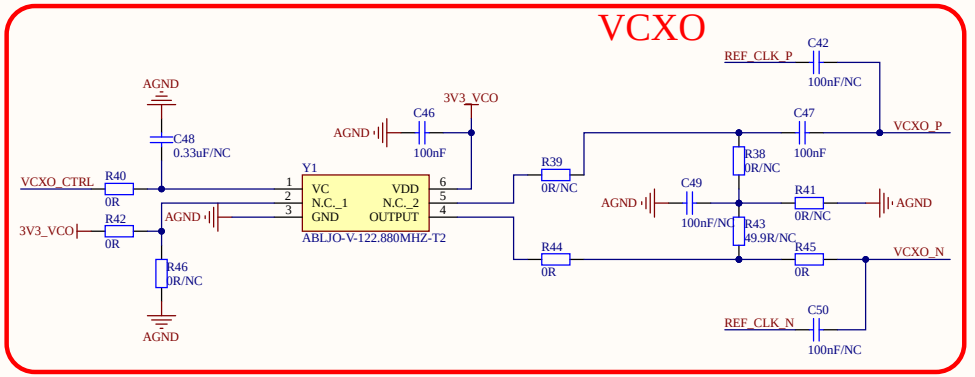



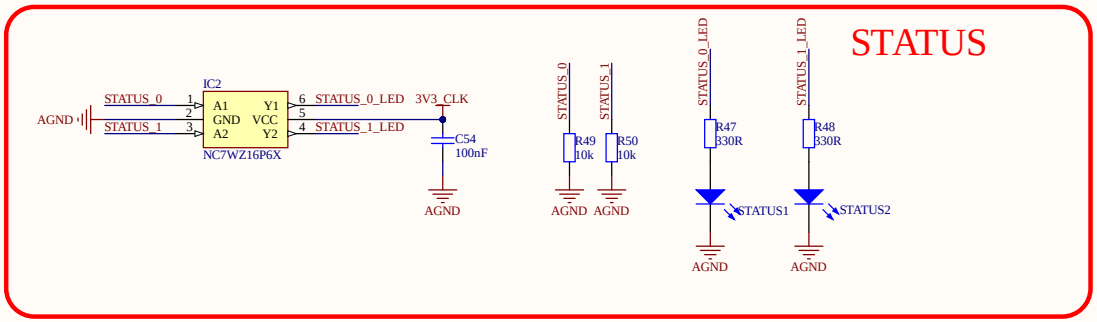



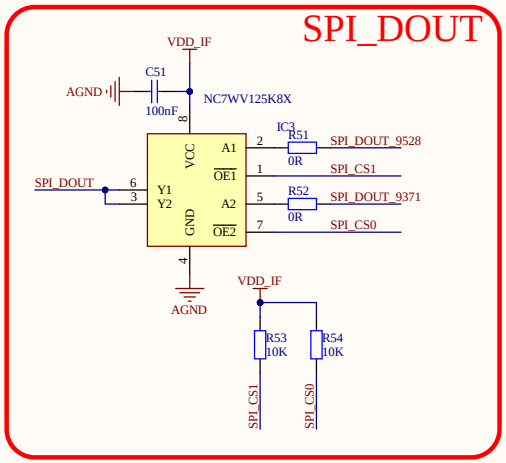

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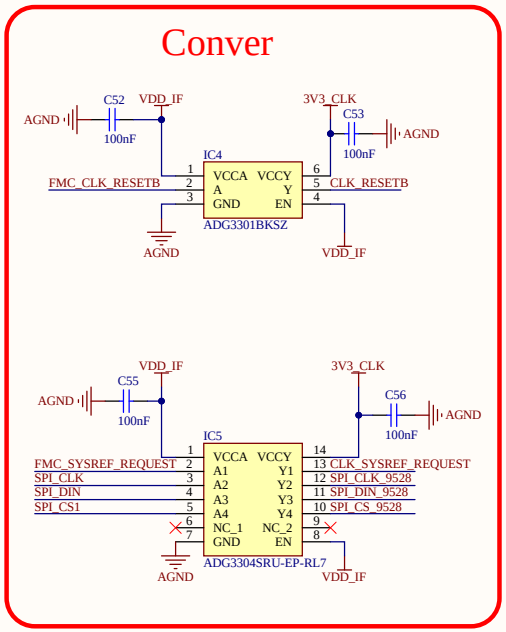


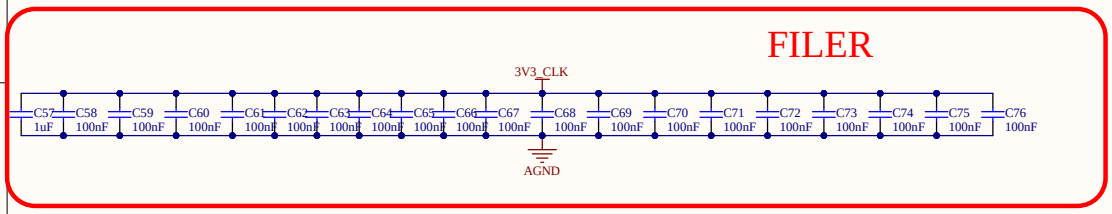



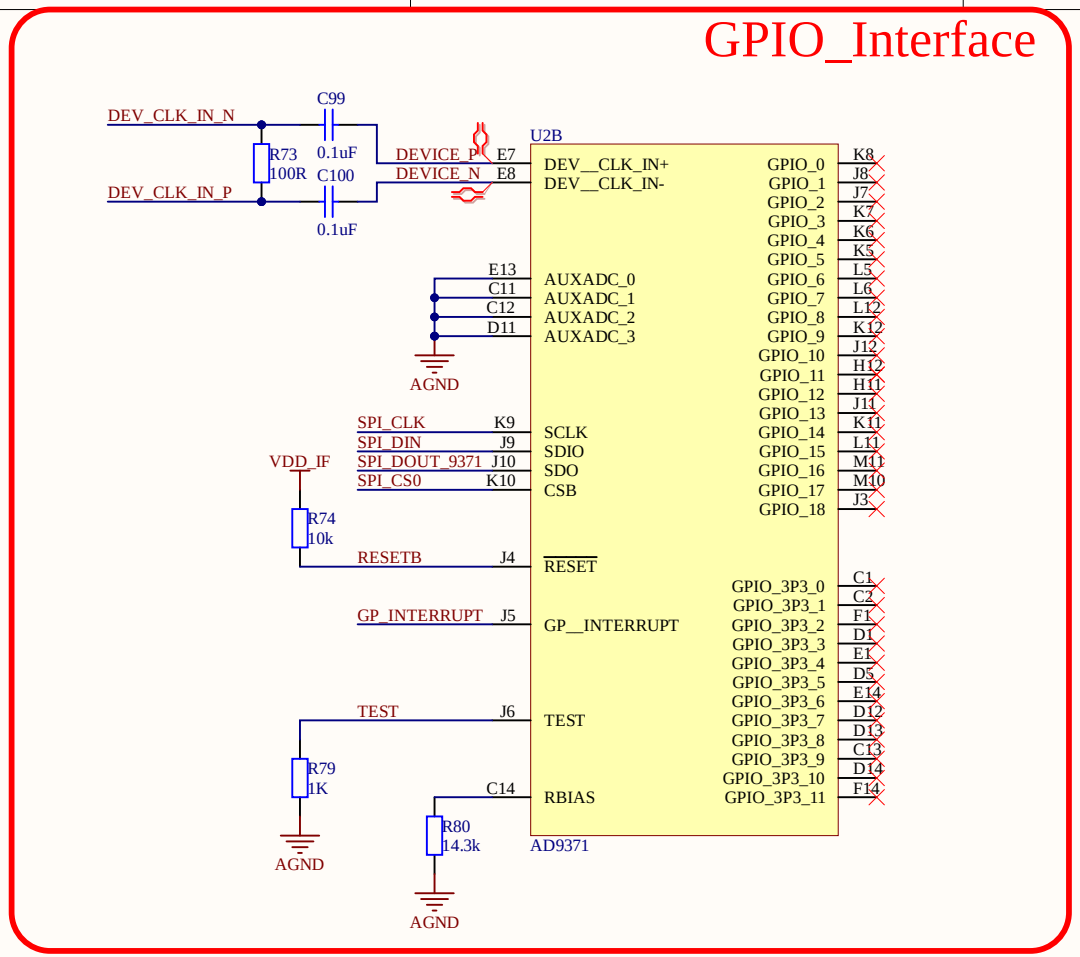



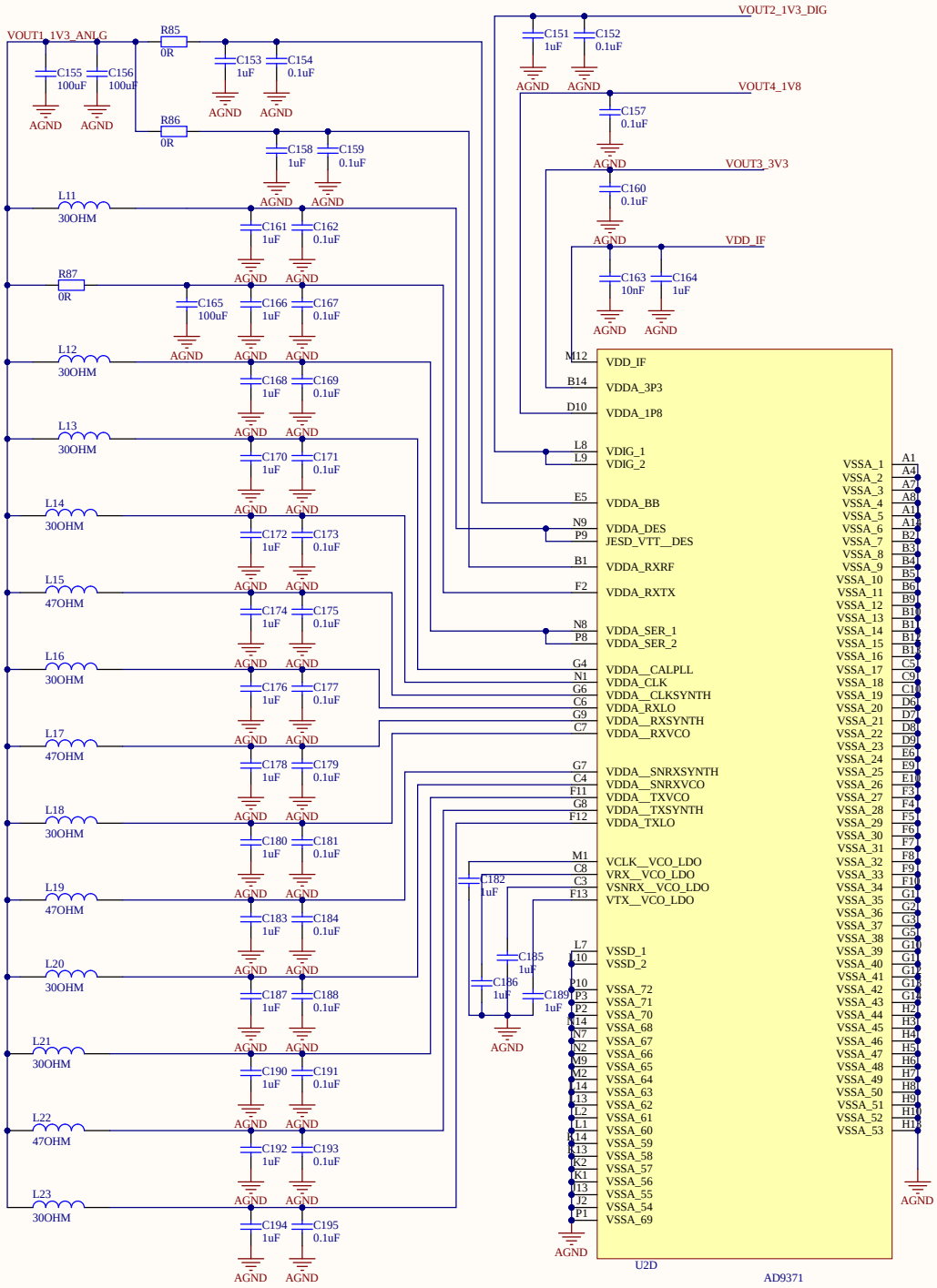



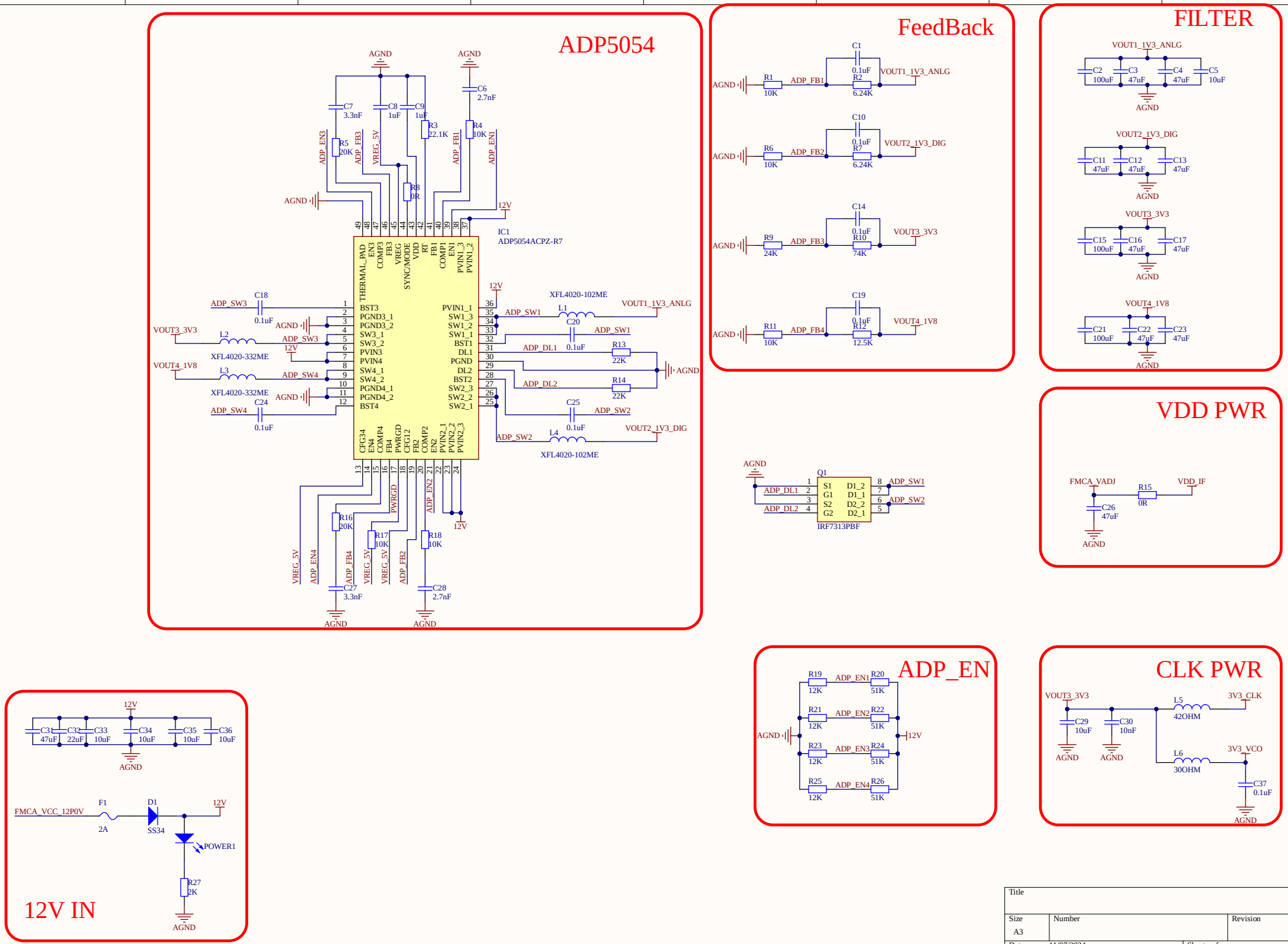

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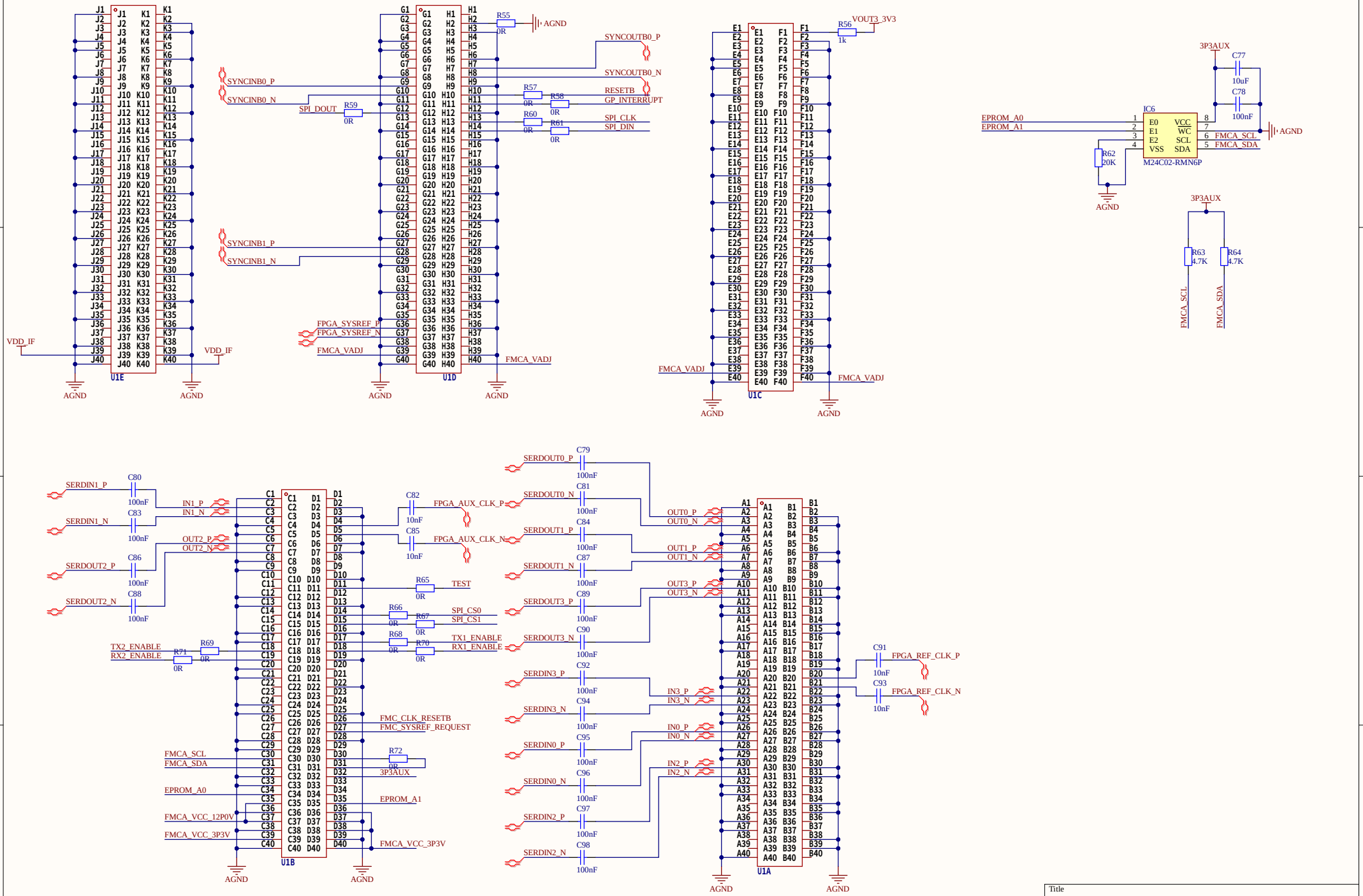



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| Date: 11/07/2024 | | Sheet of |
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ADP5054
IC1
ADP5054ACPZ-R7
THERMAL_PAD
BST3
PGND3_1
PGND3_2
SW3_1
SW3_2
PVIN3
PVIN4
SW4_1
SW4_2
PGND4_1
PGND4_2
BST4
PVIN1_1
SW1_3
SW1_2
SW1_1
BST1
DL1
PGND
DL2
BST2
SW2_3
SW2_2
SW2_1
CFG34
EN4
COMP4
FB4
PWRGD
CFG12
FB2
COMP2
EN2
PVIN2_1
PVIN2_2
PVIN2_3
EN3
COMP3
FB3
VREG
SYNC/MODE
VDD
RT
FB1
COMP1
EN1
PVIN1_3
PVIN1_2
AGND
12V
C6 2.7nF
C7 3.3nF
C8 1uF
C9 1uF
R3 22.1K
R4 10K
R5 20K
R8 0R
ADP_EN2
ADP_FB3
VREG_5V
ADP_FB1
ADP_EN1
C18 0.1uF
ADP_SW3
L2 XFL4020-332ME
VOUT3_3V3
L3 XFL4020-332ME
VOUT4_1V8
C24 0.1uF
ADP_SW4
XFL4020-102ME
L1
L4
C20 0.1uF
C25 0.1uF
ADP_SW1
ADP_SW2
VOUT1_1V3_ANLG
VOUT2_1V3_DIG
R13 22K
R14 22K
ADP_DL1
ADP_DL2
R16 20K
R17 10K
R18 10K
C27 3.3nF
C28 2.7nF
ADP_EN4
ADP_FB4
ADP_FB2
PWRGD
FeedBack
C1 0.1uF
R1 10K
R2 6.24K
ADP_FB1
VOUT1_1V3_ANLG
C10 0.1uF
R6 10K
R7 6.24K
ADP_FB2
VOUT2_1V3_DIG
C14 0.1uF
R9 24K
R10 74K
ADP_FB3
VOUT3_3V3
C19 0.1uF
R11 10K
R12 12.5K
ADP_FB4
VOUT4_1V8
FILTER
VOUT1_1V3_ANLG
C2 100uF
C3 47uF
C4 47uF
C5 10uF
VOUT2_1V3_DIG
C11 47uF
C12 47uF
C13 47uF
VOUT3_3V3
C15 100uF
C16 47uF
C17 47uF
VOUT4_1V8
C21 100uF
C22 47uF
C23 47uF
Q1
IRF7313PBF
S1
G1
S2
G2
D1_2
D1_1
D2_2
D2_1
VDD PWR
FMCA_VADJ
R15 0R
VDD_IF
C26 47uF
ADP_EN
R19 12K
R20 51K
R21 12K
R22 51K
R23 12K
R24 51K
R25 12K
R26 51K
ADP_EN1
ADP_EN2
ADP_EN3
ADP_EN4
CLK PWR
VOUT3_3V3
C29 10uF
C30 10nF
L5 42OHM
L6 30OHM
3V3_CLK
3V3_VCO
C37 0.1uF
12V IN
C31 47uF
C32 22uF
C33 10uF
C34 10uF
C35 10uF
C36 10uF
FMCA_VCC_12P0V
F1 2A
D1 SS34
POWER1
R27 2K
Title
Size A3
Number
Revision
Date: 11/07/2024
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U1E
U1D
U1C
U1B
U1A
VDD_IF
AGND
SYNCINB0_P
SYNCINB0_N
SYNCINB1_P
SYNCINB1_N
SPI_DOUT
R59
0R
R55
SYNCOUTB0_P
SYNCOUTB0_N
RESETB
GP_INTERRUPT
R57
R58
R60
R61
SPI_CLK
SPI_DIN
FPGA_SYSREF_P
FPGA_SYSREF_N
FMCA_VADJ
R56
VOUT3_3V3
1k
IC6
M24C02-RMN6P
EPROM_A0
EPROM_A1
3P3AUX
C77
10uF
C78
100nF
FMCA_SCL
FMCA_SDA
R62
20K
R63
4.7K
R64
4.7K
SERDIN1_P
SERDIN1_N
SERDOUT2_P
SERDOUT2_N
C80
C83
C86
C88
100nF
IN1_P
IN1_N
OUT2_P
OUT2_N
C82
C85
10nF
FPGA_AUX_CLK_P
FPGA_AUX_CLK_N
R65
TEST
R66
R67
SPI_CS0
SPI_CS1
R68
R70
TX1_ENABLE
RX1_ENABLE
TX2_ENABLE
RX2_ENABLE
R69
R71
FMC_CLK_RESETB
FMC_SYSREF_REQUEST
R72
FMCA_VCC_12P0V
FMCA_VCC_3P3V
SERDOUT0_P
SERDOUT0_N
SERDOUT1_P
SERDOUT1_N
SERDOUT3_P
SERDOUT3_N
SERDIN3_P
SERDIN3_N
SERDIN0_P
SERDIN0_N
SERDIN2_P
SERDIN2_N
C79
C81
C84
C87
C89
C90
C92
C94
C95
C96
C97
C98
OUT0_P
OUT0_N
OUT1_P
OUT1_N
OUT3_P
OUT3_N
IN3_P
IN3_N
IN0_P
IN0_N
IN2_P
IN2_N
C91
C93
FPGA_REF_CLK_P
FPGA_REF_CLK_N
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Size
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Sheet of
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